CS 505, Part V: Big Data - new directions in DB

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What is Big Data?[2]

- Here is a definition: *Big Data* is data that exceeds processing capabilities of conventional database systems
- Thus, resources that are needed to process such data are bigger than those offered by conventional RDBMSes and likely even so-called database appliances
- A variation on the theme of *Big Data* is *Fast Data* where the amount of storage for the data is prohibitive and for that reason we can keep only "digest" of the data

[2] The material in this part of the lecture is based on a book *Big Data Now* published by O'Reilly, and authored by a group of people.

Examples of *Big Data*

- WWW pages (an interesting case because it is *not* data that can be naturally stored in relational databases)
- Credit card transactions information (this data has a format that could conform to relational model, but the amount is too big to store that data in a conventional RDBMS)
- Social networks data (may or may not be relational, think: *Twitter*, *Tumblr*, *Facebook*, or *Reddit*)

Big Data, cont'd

- As is the case in the usual database use, *Big Data* is not kept because the businesses want to keep data, but because they want to **use** *Big Data*
- The tools that use *Big Data* are usually called *analytics*
- Examples of analytics:
  - Supermarket industry - analyzing individual shoppers habits and also performance and resupplying stores
  - Social networks - analyzing members' behavior in expectation that this information can be used to better fit the advertisers data to users' profiles
  - Big vendors: (*Amazon*, *eBay*, other large online stores,) analyze the purchase data so clients can be directed to information that may be of interest

Example of *Fast Data*

- Data on the flows in networks; even relatively small networks such as the network of UK generate the amount of data that can not really be stored
- (Storing the entire traffic of UK is beyond capabilities of our system)
- So, realistically, we can visit the flow data just once, get some information, and then discard it

Search Engines

- Original search engines, such as *Netscape* or *Alta Vista* (evolved to *Yahoo* today) did not measure relevance of the answers to the query
- It was *Google* that introduced *page ranking* (Brin and Page - who are they?)
- The idea was to find what influences relevance of pages to queries
- This is a large distributed computation involving potentially large number of pages
- These pages have to be stored, and their number grows constantly
- Thus data centers needed

Search Engines, cont’d

- To be cost-competitive, data centers use commodity processors, and commodity storage
- But these have relatively low MTTF (mean time to failure, what is it?)
- That meant that one needs to keep several copies of data
- Usually at least one in a different geolocation (why?)
- Page ranking algorithm was improved many times by Google and while the original Page algorithm is being taught in social-choice classes, the search engines such as Google, Yahoo, Opera, Bing use proprietary algorithms
- And, of course, there are special engines (such a TOR version of FireFox) that allow for anonymous searches
- This has important social consequences (why?)

Online advertising

- Search engines are a true game-changer for the society
- The issue, however, is how the search is financed (we do not pay for it)
- The results of the search are accompanied by advertisements of goods and services
- The raise of *e-commerce* resulted in the need for advertising
- How could we know whom to advertise and what?
- When you and I search, we leave a trail
- For instance my IP address was doing search for *oak floor*
- Google knows that!

Online advertising, cont'd

- So, when an add agency buys from Google the advertise space for *Hardfloor Sellers* (similarity to any real company is absolutely accidental), Google knows that this ad should be shown to that IP address when search results are served
- There is nontrivial computation related (how much charge, how long to show)
- The reason is that I will eventually buy that flooring at Hardfloor Sellers, or somewhere else and pushing it on me for long time is economically wasteful

Online advertising, cont'd

- For that reason showing these ads in pages sent to my IP address needs to be *discounted in time*
- Also, if *dexter* likes to read morbid news (for instance: *a corpse found in luggage*) then showing him ads for luggage may be a mistake
- There are also issues related to showing things together (ads for competing travel agencies?)
- *dexter* read about the MH 370 case, are we going to serve her an ad of an airline?
- Thus *semantics* of ads is an issue

Google's dilemma

- Thus it is *not* enough to store pages - additional data related to *dexter* interest (i.e. searches) needs to be kept
- But then, Google knows quite a lot about *dexter*
- What does Google does with all this information?
- How long will they store this information?
- To whom will they sell it?
- In other words, whose property is it?

Free service sites

- *Youtube* provides the ads, too
- This is not surprising, given that they are a part of Google
- Information collected by *Google* can then be used to serve *dexter* ads on *Youtube*
- But also the information gained from Youtube (*dexter* looked at the way people bake their pies) could be used by Google (show *dexter* an ad of the flour)

Social networks sites

- Sites such as *Facebook*, *LinkedIn*, *Twitter* get a significant amount of information on their members
- This can (and in fact is) used to present ads for their customers
- For that reason the size of *customer base* of such sites matter and these sites attempt to grow it, for instance by inviting individuals to join
- In the process popular services such as *WhatsApp* with a wide customer base become very valuable
- (It is not clear what will happen when a better messaging service will appear, esp. in view of recent security problems of WhatsApp)

Recommender Systems

- *e-commerce* sites also get information about the searches of their catalogues
- (For instance *Amazon*, *eBay* and smaller companies in e-commerce)
- It is in their interest to present a returning customer with *recommendations*
- For instance, for a number of times, *dexter* was looking at *minature railways*
- Then, when *dexter* visits (it is really the IP address, not *dexter* - unless *dexter* logs in), a recommendation of some of their minature railways will be shown to *dexter*
- This is related to various categories of goods
- Not only will Amazon learn about *dexter* tastes, but also it will mine the data for the entire population - customers that like minature railways like to read books involving Transybirian Railway (this is a conjecture...)

Recommender Systems, cont'd

- It should be clear that maintaining of a Recommender System requires collecting customer data
- This is similar, but not identical, to Google's collection of data on searches
- (Where is a difference?)
- The larger the organization is, esp. in e-commerce, the bigger temptation to provide the customers with recommendations
- Thus *eBay*, *Walmart* and large Department stores (*Macy's*) collect data about the customers
- The Google dilemma discussed above is still present

What can we do with the data?

- Before we deploy systems based on that data, we should *simulate* the resulting system
- So, for instance, we need to see what are possible reactions of customers to the system so deployed
- Say, we collect data on traffic in Lexington
- Let us assume a significant amount of data is collected
- If we intend to do something based on that data (for instance build a new street, or change an existing one,) then we need to simulate the result

What can we do with the data, cont'd

- If we are going to put ads on some pages, we need to see how often we are going to put these ads
- ("*If I see one more ad about Revlon again, I will drop Netflix, I swear*")
- If we are going to put a Recommender system we need to see how it will be accepted
- In short, a simulation is needed and maybe additional studies are needed
- This is not that different from general principles of deployment of a software system

What kind of CS technology used?

- Data, Big Data in particular, is useful only if we do something with it
- Once we collect data, we need to see *what does it mean*
- That is, the numbers themselves do not offer any advantage
- Their meaning (i.e. an interpretation) is what we are after
- The technology that supports the use of bigData is *Machine Learning*

Machine Learning

- Both Big Data and Fast Data are processed without human involvement
- Generally, we attempt to see statistical relationships between various data items
- These relationships are almost always statistical, not causal
- (For instance we find that individuals that like miniature railways with high probability like to dine on trains, so we show *dexter* an ad of dinner train in Bardstown, KY, but actually *dexter* hates such dinner trips)
- There are various techniques for Machine Learning

Machine Learning, cont'd

- There are numerous ML techniques. They include
  - *Supervised learning* (where we have a *labeled training set* and we have data that we categorize by similarity to this training set.) For instance hand-writing recognition is a classical example of supervised learning (we have samples of *dexter* writing, now we try to recognize more)
  - *Unsupervised learning.* Here we may want to group data into clusters (crimes; do we have a group of crimes perpetrated by same criminal?)
  - *Hidden Markov model* where we try to discover a Markov model (probabilistic transition system)

Machine Learning, cont'd

- There are more ML techniques. These include
  - *Classifiers*, for instance *Support Vector Machine* where we try to classify data into two types so that some parameter is minimized. Example of such classifier would be where we want to categorize human images into images of men and of women. Another example include discovering which luxury goods are real and which are fakes, etc.
  - (There are *many* more techniques, with many applications)

Machine Learning, cont'd

- Quite often the amount of data is too big to be handled by humans. For instance think about credit card transaction fraud
- (Here the issue is a delicate one, a transaction may be fraudulent but checking it requires some contact with the customer)
- Credit Card transaction data is stored, but a natural way to process it is via Fast Data process
- By this we mean inspecting each credit card transaction *once* as it comes in, make some decision
- (Later, if needed, revisiting)
- Quite often we talk about *streaming data* in similar situations

Machine Learning applications

- The issue is *when* one should think about ML applications
- A precondition, of course, is that data is available
- If data can be inspected only *once* before storing or maybe discarding) then pretty complex processing can be done
- But also think that *before* it has been discovered that CC numbers and additional data were stolen from *Target* credit card companies knew that *likely Target* was the source of stolen cards
- How did they know it?

ML applications, cont'd

- Using ML requires working knowledge of statistics
- (Otherwise, it is like in a well-known dictum of B. Disraeli: "Lies, big lies, and Statistics")
- For that reason *before* deploying ML applications extensive studies of data are needed
- ("Do we collect the right data?")

ML applications, cont'd

- Target breach was *not* discovered by streaming but by learning that some fraudulent CC transactions were *similar* to *Target* CC transactions
- This must have happened by ML, not by a single inspection like in streaming
- (Of course, *afterwards* streaming inspection could be used, why?)
- Can you think about other ML applications?

Biomedical applications of Big Data

- DNA databases store fragments of DNA (that is their linguistic representations)
- Data mining of these allows for recognition relationships between medical conditions and DNA fragments properties
- This is often called *Genomics* (What is Genomics?)
- Genes are *expressed* and the expression of genes leads to creations of *proteins*
- At present biologists talk abouts *Proteomics* (a mixture of *protein* and *genome*)
- (What are differences between genes and proteins?)

Biomedical applications cont'd

- But there is more; one can look at relationship with enzymes and other components of cells
- (What is *enzymatics*?)
- One can try to breed specific features in animals (like more meat in turkeys) by use genetic information
- Another issue (not solved as of today) is the question of biomedical information. There are too many biomedical publications (even worthy biomedical publications) to be read by a researcher. The issue is how to help researchers (and MDs) to get to relevant literature

Biomedical applications cont'd

- Clearly, if *some* documents are labeled then we can use ML techniques to label other documents
- But this is *dynamic*; we can later discover that other labels are applicable
- Biomedical literature has all sorts of problems (for instance, what would one do if someone claims that somebody else's results can not be confirmed?)
- More generally, what to do with *retractions*?
- How to handle changing interests of users?
- What to do with claims of *drug interactions*? After all it is a special category, but extremely important (ever heard about *Iatrogenic* effects?)

Biomedical applications cont'd

- There is a system called *PubMed*, maintained by *National Library of Medicine* (a part of NIH)
- This database (and an interface, with many *search functions*) provides biological and medical information. At presence it contains over 23 million articles
- (What does it mean for the users?)
- PubMed maintains a database of medical terms, called *MeSH* (*Medical Subjects Headings*)
- (And there are other tools, for instance *Universal Medical Language System*, ULMS), and yet another terminology system (basically for pathologists) called *SNOMED*)
- Observe that all these efforts are *culture dependent*, (why?)

Astrophysicists

- A number of orbiting satellites provides a constant stream of geophysical and astronomical data
- For instance *Orbiting Carbon Observatory* collects and transmits to Earth the data about the carbon dioxide content of atmosphere
- The most spectacular was *Hubble orbiting telescope* launched from the Shuttle *Discovery* in the 1990
- (And do you know what *Kepler Mission* is?)
- (Discovering planets with a potential for sustaining life)
- (Some were found)
- There are other orbiting telescopes observing other parts of the electromagnetic spectrum (for instance X-ray telescopes)

Astrophysicists, cont'd

- Data so collected is, of course, stored, analyzed and visualized in a variety of ways
- Astrophysicists (and more generally other *hard* scientists) face a problem similar to that of biologists/MDs - they need to keep current on literature, and they need to *mine* their literature
- The resulting NASA *Astrophysics Data System* provides the access to the abstracts and full text papers and other items (circa 10.5 million) and is hosted by Harvard-Smithsonian Center for Astrophysics
- They also provide an API that can be used for *Do-it-yourself* research
- (What kind of benefits astrophysicists and, more generally, society gets out of such repositories?)

Big data in Geographical IS

- While the Earth seems very small from cosmic perspective, for humans it is a rather big place
- Since the time of European expansion (ca. 1500) maps were made
- But there is plenty under our feet, and in fact in many layers
- Since 1980ies *Geographic Information Systems* (GIS)
- 1986 - MIDAS - Mapping Display and Analysis System
- Generally, GIS contains many layers of information (surface, geology, hydrology and others - for instance density of population)
- As you travel US by car and collect state maps, you should recognize that they contains various classes of geographical information (which ones?)

Applications in Geology

- Humanity mines considerable resources from under the surface of Earth
- Think about coal (at present the most important source of energy)
- There are many other minerals being mined (what?)
- Our technology requires (at present) availability of various minerals (ever heard about *Rear Earths*?)
- Our phones would not work without them ...
- More generally (because it is not only about *Lanthanides*) we need information about natural resources
- We also need information about *geological formations*

Applications in Geology, cont'd

- And, of course, do not forget about oil and natural gas
- As of today, our society can not function without them
- So, our GISes need the information about all these resources, and there should be large databases that contain such information
- In US, such databases are maintained by *US Geological Survey* - a government organization dealing with geology, and more generally, geography (they also deal with *seismic information*)
- (Surely there must be other such databases, maintained by oil companies, gas companies, and mining companies)

Applications in Geology, cont'd

- There is a *National Geologic Map Database* (http://ngmdb.usgs.gov)
- As in case of other large databases the issue of the terminology and the *meaning* of terms arises
- So, they also have *GEOLEX*, the lexicon of terms used in the maps
- There are specific problems related to the fact that they deal with a three-dimensional structure
- (This is *stratigraphy* (what is it?) and its nomenclature)
- The information changes (where did we see information change before?) and keeping current such databases is a challenge
- The "big picture" - satellite view is, of course, an issue

Applications in Geology, cont'd

- Kentucky has its own Geological Survey (why?)
- It happens to be located at UK (see http://kgs.uky.edu)
- All sort of resources, maps of minerals but also of underground water resources are available there
- Other states and territories all have similar institutions, but names vary

COMPENDEX

- *COMPENDEX* is a large database related to engineering information (the name is *Computerized Engineering Index*)
- It is commercial, not freely available (recall that PubMed was free)
- It is maintained by the publishing company *Elsevier*
- Has over 16 million records (of various sorts) and covers entire engineering
- It is an inheritor to *Engineering Index*, an institution 125 years old
- It has a number of additional database used by engineers in design and implementation

Problems with Big Data

- Generally, in Big Data area we deal with the three kinds of issues
  - Volume
  - Velocity
  - Variety
- This is often called that data is *big*, *fast*, and *varied*
- The anecdotal evidence is that in a non-cloud environment, any *two* of these issues can be handled together but not all three

Problems with Big Data, cont'd

- *Google* (and their competitors) had to deal with all three issues at once (why?) and found solution (for their problems) in tools like *Bigtable*, data centers, commodity storage and processing
- They process data repeatedly (because, for instance, data about *dexter* changes frequently) and can not depend on pre-processed answers to queries
- This has consequences (which ones?)
- In particular repeated queries may produce different results

Can Big Data be a social issue?

- All of us leave a *data trail*, for instance by using credit cards
- Could this be used against us?
- For instance could a mortgage loan be affected by the fact you shop in stores generally considered *lower end*?
- (Actually, an anecdotal information points to a CC company lowering credit card limit on somebody's CC upon such event)
- OKCupid case: finding correlation between the race and what people like (what is OKCupid, and why that information was even known?)
- Going backwards: white people like hamburgers, *marek* likes hamburgers, so *marek is white*
- Various social networks get the profile data of users

...

What can be done about Big Data?

- Many companies (and Government on the top of it) collect data about us
- It is quite difficult to control how this data is used
- Companies sell data about customers
- That data can be aggregated and mined
- A lot about *dexter* can be learned in the process
- In spite of various rules there are no guarantees of privacy *and can not be* because data may lead to unexpected conclusions. For instance *marek* happens to have a parakeet, the only one in *Bardstown*. And someone in Bardstown bought both parakeet food, and *caucasian ovcharka muzzle*
- So we can learn about *marek's dog*

BD and Social Networks

- Companies need to check the information about themselves that appears on social networks
- For instance if you are in business of selling yarn, you need to know what people say on *Ravelry*
- (And *Michelob* needs to know what customers say about *Michelob Ultra* on *Untappd*)
- This information is used by Customer Service, but also by Marketing (*We need to advertise more, people do not talk enough about Michelob Ultra*)
- This grows new uses of social networks (specialized networks discussed above, but also *Facebook* and *Twitter*)

Abuse of Big Data

- Really, it is about abuse of Statistics
- Here are findings:
  - The more advanced the statistical methods used, the fewer critics are available to be properly skeptical (in other words people will focus what they understand, leaving things they do not understand to experts)
  - The more advanced the statistical methods used, the more likely the data analyst will be to use math as a shield (You think: *I do not understand the math, so likely it is right...*)
  - Any sufficiently advanced statistics can trick people ito believing the results reflect truth
  - And, of course, correlation does not mean causation (diabetes often goes together with obesity, hence diabetes causes obesity)

Healthcare on national scale

- Healthcare is a significant part of national spending
- The issue is how BD could contribute to streamlining medical costs
- Generally, medicine operates *today* by means of *standard of care* concept
- (When you have a medical problem then there are standard techniques to handle such problem)

Healthcare on national scale, cont'd

- Standard of care is based on large standardized studies
- For instance *dexter* has a bacterial respiratory tract infection. Then the standard of care requires the medical practitioner (MP) checks *dexter*, asks for tests (maybe *dexter* has viral RTI, then it is different), prescribes one of antibiotics (for which *dexter* is not allergic), discusses palliatives (taking care of cough, fever, running nose) and sends *dexter* to bed for 3 days
- This is based on large studies and statistical properties of the population

What about more complex problems?

- Let us discuss one of very significant medical problems, namely *breast cancer*
- It is known that breast cancer is not a single disease, but one that is related to a number of genetic variations (mostly at women)
- Generally, it is known that a well-established drug, called *tamoxifen* works (after the operation and removal of primary tumor) in 80% of cases
- But it is also known that if the patient has a gene called *her2* then another drug, called *herceptin* is significantly more effective

Complex problems cont'd

- We can now find the DNA sequence of the patient (a lot of data, but now it is doable for less than $10,000, maybe significantly less and soon for *much* less)
- Now, we can check the DNA of the presence for subsequence that expresses itself as *her2* gene
- This is a BD task
- This is an example of cost-effective, personalized medicine driven by BD
- Of course, breast cancer is not the only case where a personalized medicine becomes a reality

Personalized medicine and Big Data

- What can Big Data do to help patients?
- Our example above shows that Big Data has potential to significantly improve standard of care by introducing *deep*, i.e. not easily visible patient-specific information
- But how does it fit computers?
- There must be standards for encoding information so that it is computer readible, i.e. computerized patient system record standard
- There are various standards for CPRS
- US Veterans administration has such standard. Its *description* has 266 pages ...
- (Kind of difficult to enforce such standard, why?)

Medical Social Networks

- We have seen that social networks (*LinkedIn*, *Facebook*, *Twitter*, *Tumblr*, and others) create communities of users organized around *some* common purpose (learning about business opportunities, providing a channel for sharing news, a fast notification of news, etc.)
- This phenomenon occurs in medicine, too. There, support groups for individuals sharing the same medical problem are common
- Here is a number of networks related to medical problems:
  - *PatientsLikeMe* (which is a forum for sharing the information about specific medical problems both for patients and caregivers)
  - This network generates all sort of information about medical problems and sells that information both to medical institutions and to drug companies (but is very open about it)
  - Has > 500 groups related to medical conditions

Medical Social Networks, cont'd

- Association of Cancer Online Resources (ACOR) (http://acor.org)
  - This organization is not-for-profit, and appears to be supported by grants
  - Has over 100 groups (called mailing lists) related to specific medical problems
- *SugarStats* a social network of Diabetics patients
  - Supports diabetics patients
  - Helps to maintain information on the blood sugar levels (visualize)
  - It appears it is financed by various levels of service

But there is more ...

- There are *many* more medically motivated social networks
  - These include: *MedHelp* organized around *forums*. This is the largest online medical network, with MDs involved (http://www.medhelp.org)
  - *Disaboom*, a community of disabled individuals (for profit, supported by ads)
  - *CureTogether* provides ratings of treatments, supported by ads
  - *Vitals* - helps to find a medical specialist (sells your data to Google for choice of ads)
  - And many more

Tools for Big Data

- There are many suits of tools, both proprietary and open source
- For instance, *Google Prediction API* is a paid service (free for 6 months ...)
- Google advertises it as a tool for doing data mining in many areas, for instance:
  - Spam Detection and reputation filtering
  - Suspicious Activity Detection
  - Document and email classification
  - Customer sentiment analysis
- (And many other areas of analysis)

But there are open source tools, too

- (In 2010, Google described its internal project called *Google Dremel*)
- *Apache Drill* is an open source reimplementation of *Dremel*
- From their description: *Drill is a distributed system for interactive analysis of large-scale datasets. Drill is similar to Google's Dremel, with the additional flexibility needed to support a broader range of query languages, data formats and data sources. It is designed to efficiently process nested data. It is a design goal to scale to 10,000 servers or more and to be able to process petabyes of data and trillions of records in seconds*
- *Drill* has: flexible data formats (in particular nested format), an SQL-like query language (and pluggable query languages), and generation of fast query plans,

What did we learn about Big Data?

- It is now possible to process immense collections of Data, really on petabyte scale
- This has plenty of consequences in all sorts of ares important for modern society
- Big internet companies (*Google*, *Yahoo*, *Facebook*, *Amazon*) lead the way, not academia
- Many tools (often based on Machine Learning) form basis for the Big Data *analytics*